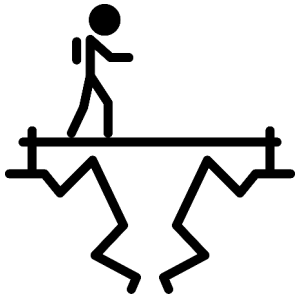
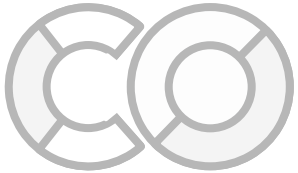
Keyword Spotting Challenges/Constraints

What are we going to learn?



Challenges with Keyword Spotting

The Keyword Spotting ML **Pipeline**



Hands-on training of a Keyword Spotting Model

Challenges and **Constraints**



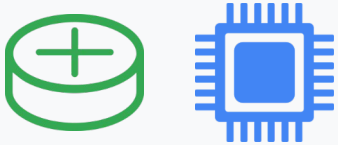
Latency & Bandwidth



Accuracy & Personalization



Security & Privacy



Battery & Memory

Challenges and **Constraints**



Latency & Bandwidth

Accuracy & Personalization

Security & Privacy

Battery & Memory

LATENCY

Provide results quickly, respond in real-time to the user

Challenges and **Constraints**



Latency & Bandwidth

Accuracy & Personalization

Security & Privacy

Battery & Memory

BANDWIDTH

Minimize data sent over the network (slow and expensive)

Challenges and **Constraints**

Latency & Bandwidth



Accuracy & Personalization

Security & Privacy

Battery & Memory

ACCURACY

Listen continuously, but only trigger at the right time

Challenges and **Constraints**

Latency & Bandwidth



Accuracy & Personalization

Security & Privacy

Battery & Memory

PERSONALIZATION

Trigger for the user and **not** for background noise

Challenges and **Constraints**

Latency & Bandwidth

Accuracy & Personalization



Security & Privacy

Battery & Memory

SECURITY

Safeguarding the data that is being sent to the cloud

Challenges and **Constraints**

Latency & Bandwidth

Accuracy & Personalization



Security & Privacy

Battery & Memory

PRIVACY

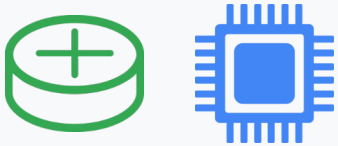
Safeguarding the data that is being sent to the cloud

Challenges and **Constraints**

Latency & Bandwidth

Accuracy & Personalization

Security & Privacy



Battery & Memory

BATTERY

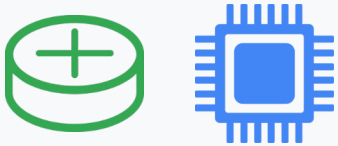
Limited energy, operate on coin-cell type batteries

Challenges and **Constraints**

Latency & Bandwidth

Accuracy & Personalization

Security & Privacy



Battery & Memory

MEMORY

Run on resource constrained devices

Challenges and **Constraints**

Latency & Bandwidth

Accuracy & Personalization

Security & Privacy

Battery & Memory

\+ ***MORE***

further constraints

Challenges and **Constraints**

Latency & Bandwidth

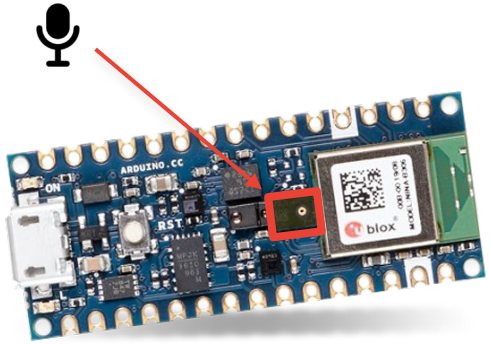
Accuracy & Personalization

Security & Privacy

Battery & Memory

So **how** do companies do ***Keyword Spotting*** today?

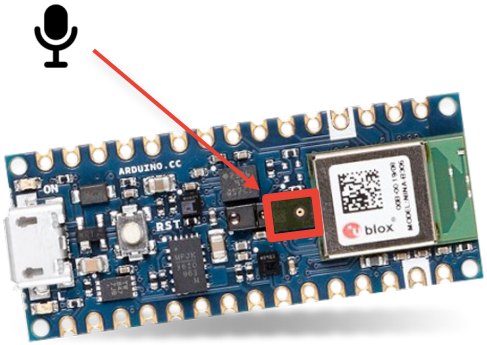
Anatomy of a Keyword Spotting Application

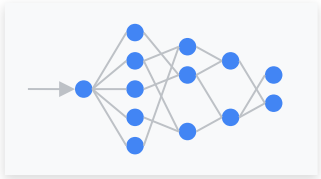


1 Continuously listen on the microcontroller

Anatomy of a Keyword Spotting Application

2 Process the data with **TinyML** at the edge

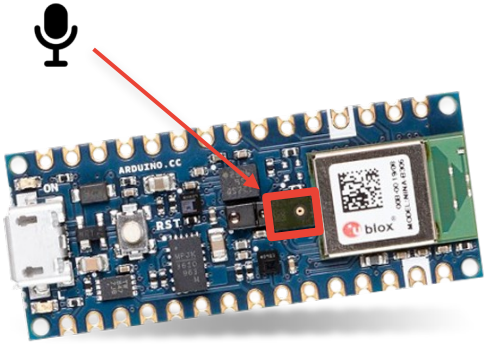


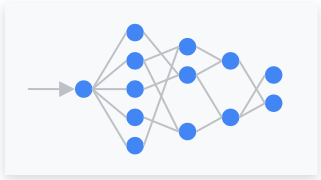


1 Continuously listen on the microcontroller

Anatomy of a Keyword Spotting Application

2 Process the data with **TinyML** at the edge







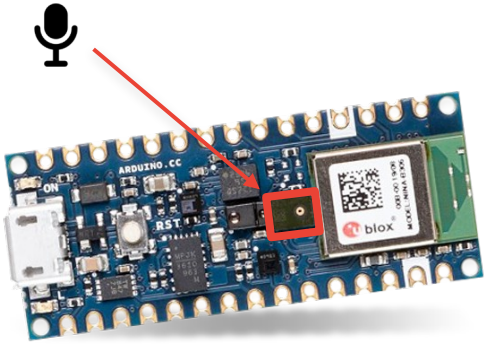
3 Send the data to the cloud when triggered

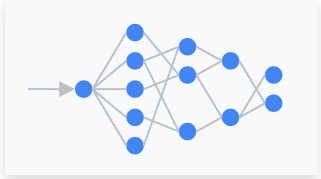
1 Continuously listen on the microcontroller

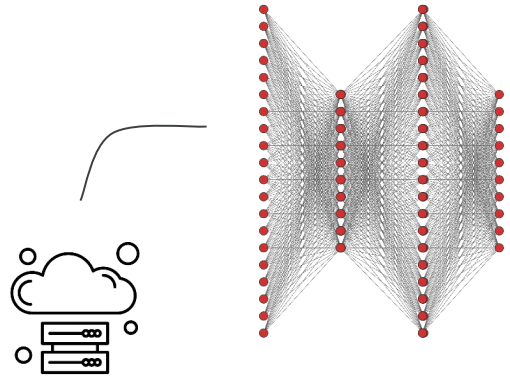
Anatomy of a Keyword Spotting Application

4 Process the full speech data with a large model in the cloud

2 Process the data with **TinyML** at the edge





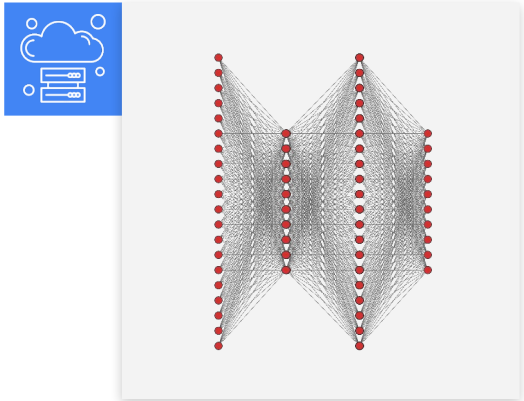


3 Send the data to the cloud when triggered

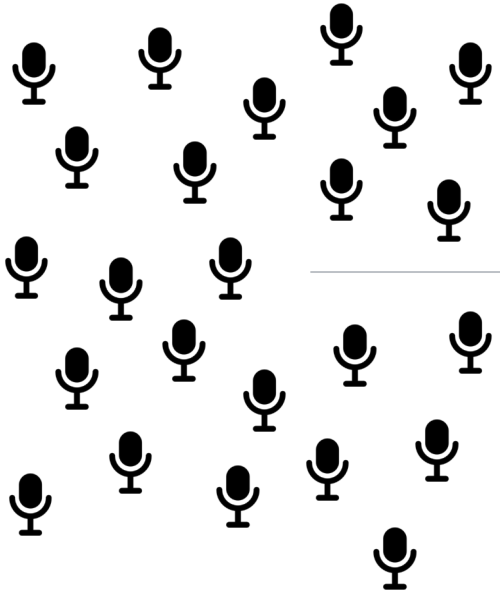
1 Continuously listen on the microcontroller

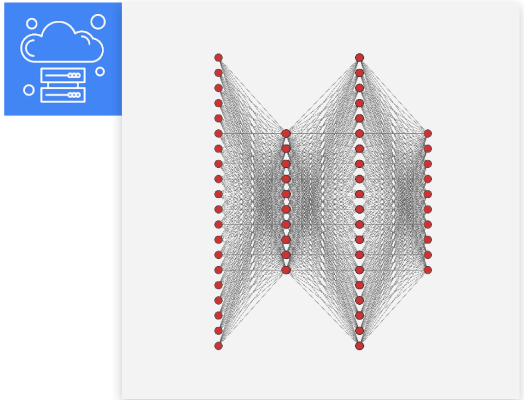
Anatomy of a Keyword Spotting Application



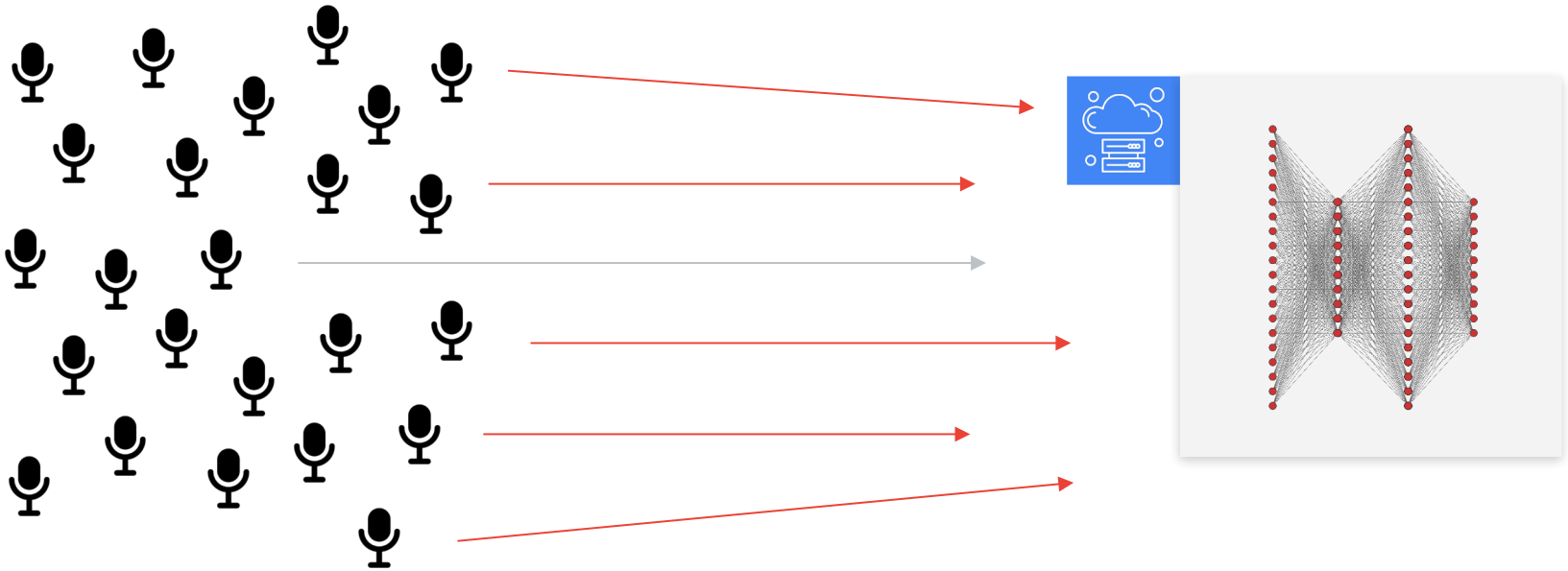


Anatomy of a Keyword Spotting Application





Anatomy of a Keyword Spotting Application



How do we scale to **billions** of users?



Source: Google

Cloud TPU



Source: Google

300 Watts



Source: Google

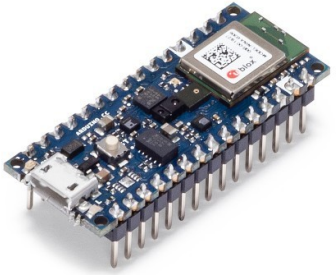
300 Watts

Cloud TPU

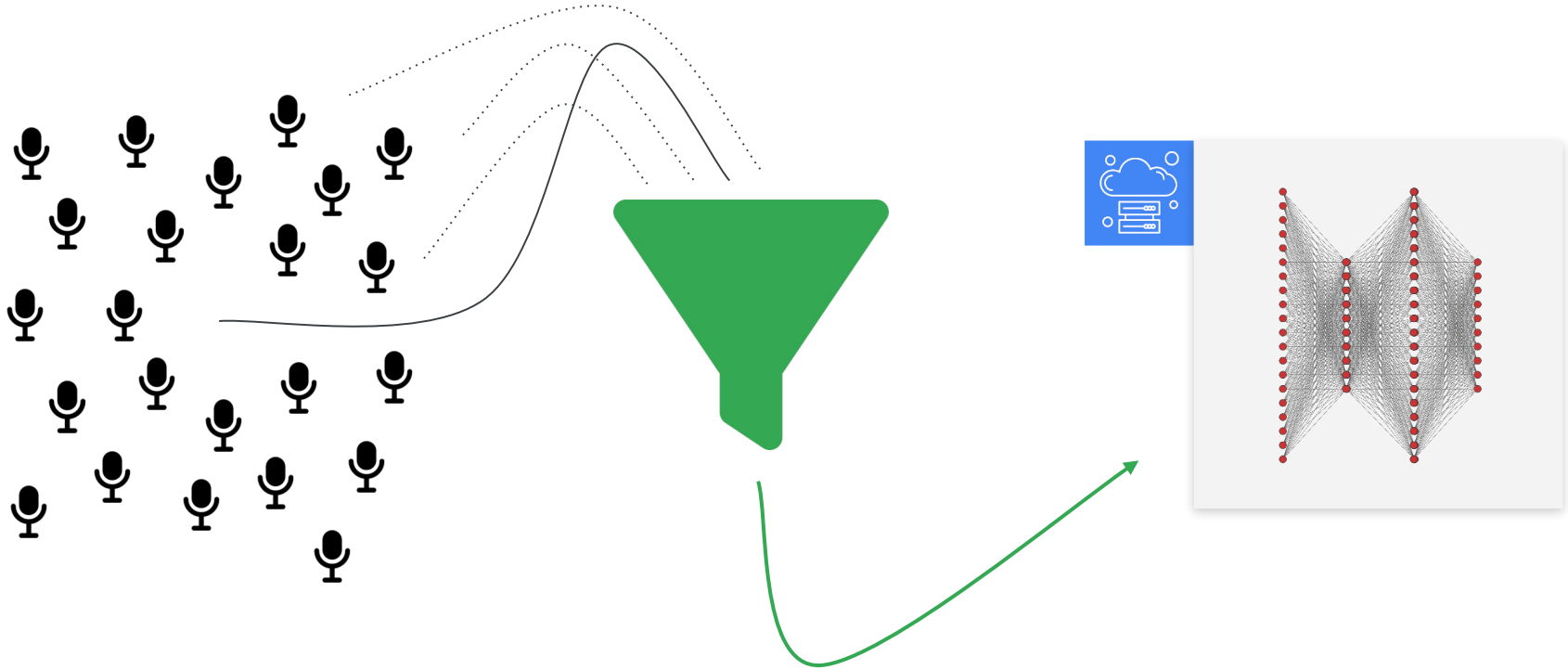


< 1 mWatt

TinyML



Anatomy of a Keyword Spotting Application



Fullscreen
Show Presenter